

Hook, problem, purpose: the Introduction

Turning a pattern on a map into a reason to read on

YOU LEAVE WITH

A draft purpose statement, a 'however' hook and a working title for a question you might investigate.



Tutorial 3 of 8 • Semester 1

Today's plan

Goal: Draft the opening and the purpose statement for a question you might investigate.

- 1 Check-in: what did your reverse outline show?
- 2 Methods: thresholds, ENSO's calendar, composites
- 3 Predict before you plot
- 4 The three moves of an Introduction; openings that work
- 5 The purpose statement; synthesising literature

BRING / HAVE READY

- Task 2A (≤ 200 words) and Task 2B (reverse outline)
- Your composite and correlation maps (notebook \$2)
- 1–2 patterns that caught your eye



Check-in: what did the reverse outline show?

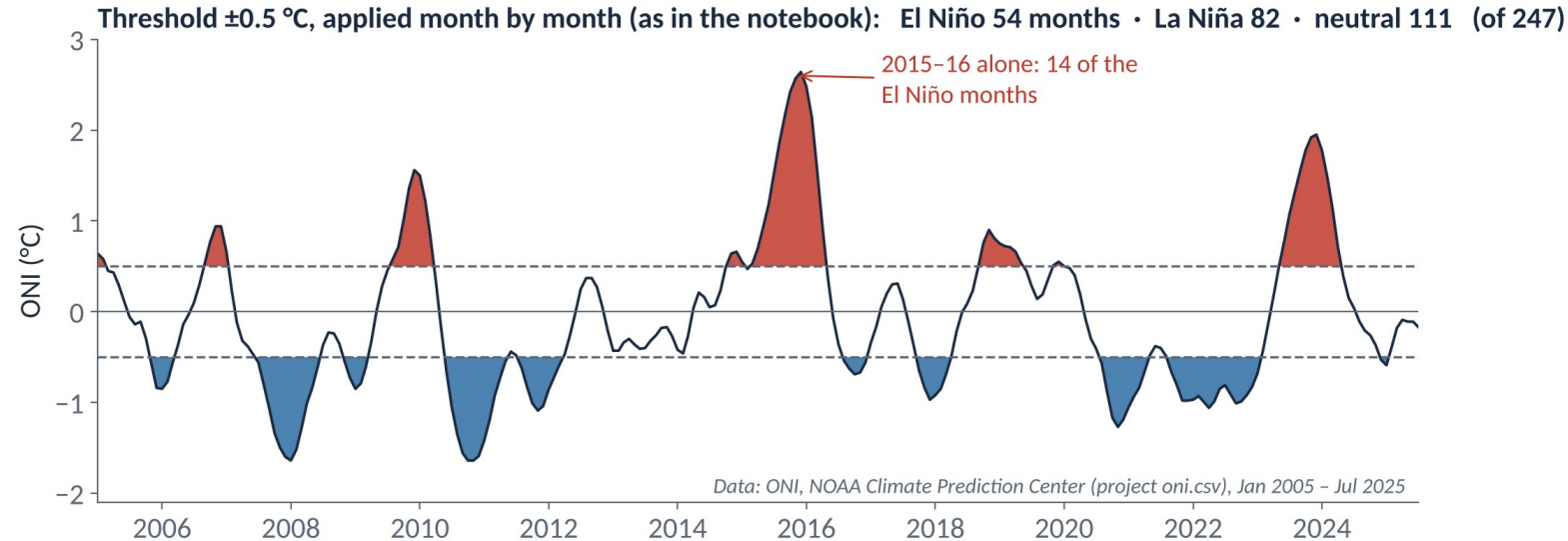
- With a neighbour (3 min), compare your reverse outlines:
- How many paragraphs before the **problem** appeared?
- Was there a clear **purpose** sentence? Where?
- Which sentence made you want to keep reading, and why?
- Then (2 min): two or three pairs report back one thing they noticed.

WHAT YOU'LL PROBABLY FIND

- Context first, kept short
- A turn ('however', 'but', 'yet', 'remains unclear')
- A purpose near the **end** of the Introduction
- Very little about the results themselves

METHODS

Choosing El Niño and La Niña months: thresholds



- **ONI:** 3-month running mean of sea-surface-temperature anomaly in the Niño 3.4 region (5°N – 5°S , 170°W – 120°W), NOAA CPC.
- **The notebook's rule:** each **month** with $\text{ONI} \geq +0.5^\circ\text{C}$ is an El Niño month, $\leq -0.5^\circ\text{C}$ La Niña, otherwise neutral (ENSO_THRESH).
- **NOAA's official episodes are stricter:** $\pm 0.5^\circ\text{C}$ must hold for ≥ 5 consecutive overlapping seasons. Say which rule you used.
- **One event dominates:** 2015–16 supplies 14 of the 54 El Niño months. Your El Niño composite is partly a 2015–16 composite.

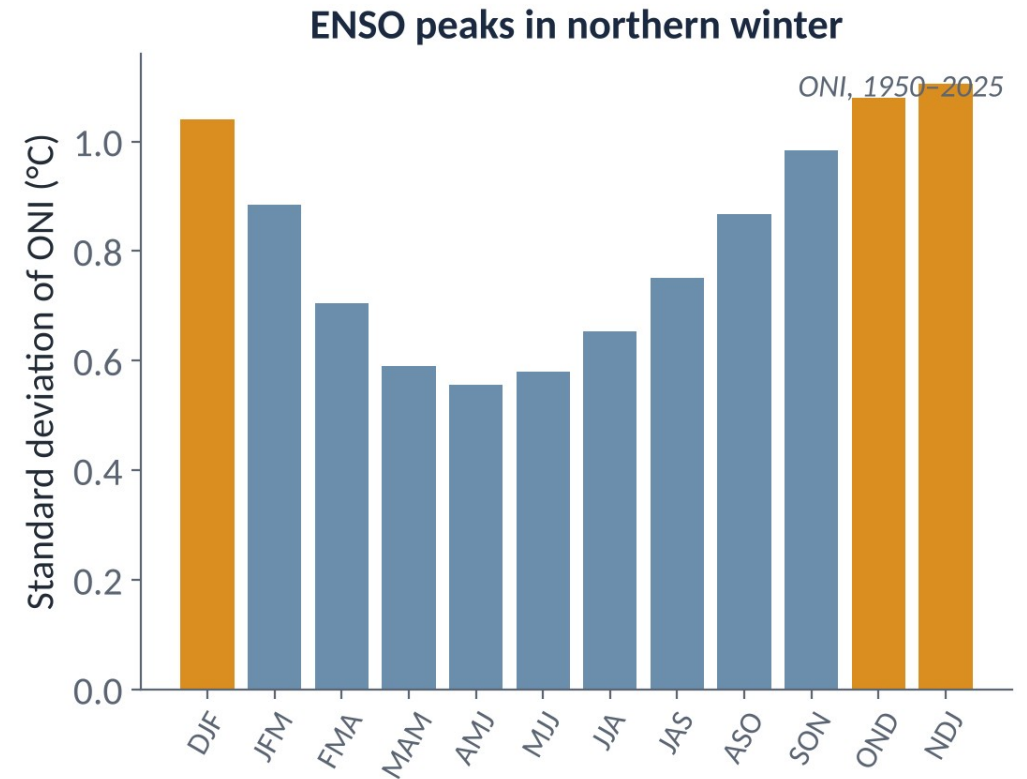
IN YOUR METHODS, THIS BECOMES

"Months with $\text{ONI} \geq 0.5^\circ\text{C}$ ($\leq -0.5^\circ\text{C}$) were classed as El Niño (La Niña), giving 54 El Niño and 82 La Niña months."

METHODS

ENSO keeps to a calendar

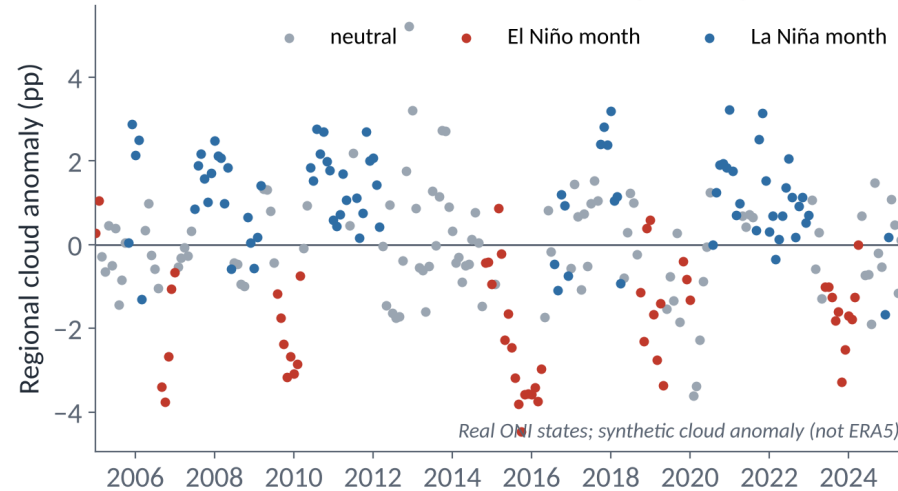
- ENSO is **phase-locked** to the seasons: events grow through northern summer and autumn, peak around **Nov–Jan**, and decay by spring.
- ONI varies about **twice** as much in Nov–Jan as in Apr–Jun (s.d. ≈ 1.1 vs 0.56 °C, 1950–2025).
- So many El Niño months are winter months. Anomalies remove the seasonal cycle, but the response itself can depend on season.
- A season-specific question (e.g. DJF only) can be a good one; notebook §6 compares DJF and JJA.
- The 3-month running mean blurs timing by about ± 1 month.



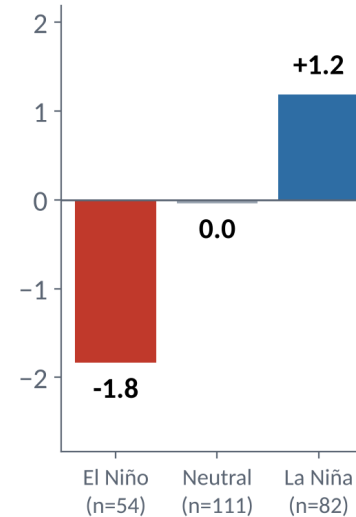
METHODS

What a composite is: El Niño – La Niña

1 Anomalies, coloured by ENSO state → 2 average each group → 3 subtract



El Niño – La Niña = -3.0 pp



- **A composite asks:** on average, how different is the anomaly in El Niño months from La Niña months?
- 1 anomalies → 2 average each group → 3 subtract. Units stay those of the variable (pp).
- Positive El Niño – La Niña = higher during El Niño. It's a difference between two states, **not** a correlation.
- Always give the **group sizes** (54 and 82 months), and look at each phase against neutral too (Tutorial 6).

IN YOUR METHODS, THIS BECOMES

“The composite is the mean anomaly over El Niño months minus that over La Niña months.”

IN-CLASS ACTIVITY • 3 MIN

Predict before you plot

- 1 Without looking at your maps, predict the sign (+, −, 0) of the El Niño – La Niña **total cloud** composite in: **(a)** region 4, Niño 4 (central Pacific); **(b)** region 5, Maritime Continent; **(c)** region 6, SE Pacific stratocumulus.
- 2 Write a **one-line physical reason** for each (Walker circulation? SST? subsidence?).
- 3 Now look at the real map (next slide) or your §2 map. Where were you wrong?

YOU LEAVE WITH

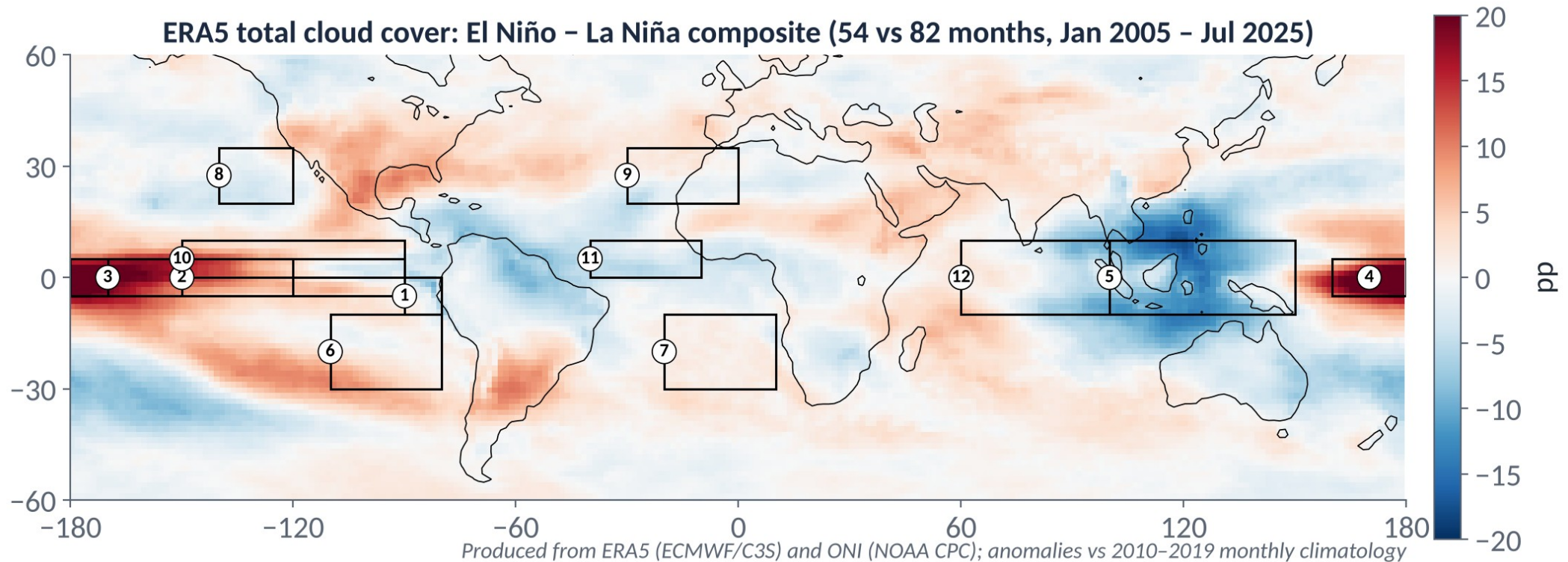
- Three predictions with reasons
- Any surprise is a candidate ‘**however**’ for your Introduction

PAIR CHECK

- Compare predictions before the reveal. Did you use the same reasoning?

METHODS

The answer: El Niño – La Niña, real ERA5 total cloud



- Strong positive over the central and eastern equatorial Pacific (regions 2–4); strong negative over the Maritime Continent (5). This is the **Walker circulation weakening** and convection shifting east.
- Weaker, patchier signals elsewhere, including the subtropical stratocumulus boxes (6–9). Some surprise people; that's a question waiting to be asked.

Every good Introduction makes three moves

IN AN ENSO-CLOUD REPORT, FOR EXAMPLE

1 CONTEXT

Common ground the reader already accepts

ENSO reorganises tropical convection through the Walker circulation, and cloud responds.

2 PROBLEM

The hook: a contrast, puzzle, gap or stake

However, cloud at different levels is controlled by different processes, so the response need not be the same everywhere.

3 RESPONSE

What this report does, and its scope

The purpose of this report is to compare ... over ... using ERA5 and the ONI.

First sentences: avoid the ‘resounding banality’

Medawar's phrase (via Schultz) for an opening that is true, obvious, and does no work at all.

WEAK

“ENSO is a very important phenomenon that affects weather all over the world. Clouds are also very important for the climate system.”

STRONG

“ENSO is the largest source of year-to-year climate variability on Earth, and clouds record it: as the Walker circulation weakens during El Niño, convection over the Maritime Continent is suppressed while parts of the central Pacific become cloudier.”

WHAT CHANGED

- **Specific:** names the mechanism (Walker circulation) and real places, instead of ‘very important’.
- **Points somewhere:** the reader can already sense the question coming (where else does cloud change, and why?).
- Schultz: avoid statements nearly everyone knows are true, **unless you are about to contradict them.**

The purpose statement: a little formula helps

- Schultz recommends every Introduction contain a sentence like “**The purpose of this paper is to ...**”
- It forces you to compress your goal into one or two sentences. If you can't, the question isn't ready yet.
- Put it in the **last paragraph** of the Introduction, after the hook.
- State the **scope**: what the report will **not** cover. That stops a reader expecting more than 2,000 words can give.
- A **hypothesis** is optional but strong: “We expect ... because ...”.

TEMPLATE

The purpose of this report is to...

- **verb**: quantify / compare / test whether ...
- **what**: the ENSO-related change in [cloud variable]
- **where/when**: over [region], in [season / all months], [period]
- **how**: using ERA5 monthly anomalies and the ONI (composites / correlation)
- **optional**: We hypothesise that ... because ...

IN-CLASS ACTIVITY • 10 MIN

From a pattern to a purpose

- 1 Pick **one or two** patterns from your composite or correlation maps that you might investigate.
- 2 For each, write a **nutshell**: one sentence that starts “The purpose of this report is to...”.
- 3 Write a **two-sentence hook**: one sentence of common ground, then one starting “However, ...”.
- 4 Write a **working title** that passes at least three of the five title tests.
- 5 Read **only your purpose sentence** to your partner. They tell you back what you will do. If they can't, rewrite it.

YOU LEAVE WITH

- 1–2 nutshells, a hook and a title in your workbook (p. for Tutorial 3)
- A first sense of which question you'll commit to on **29 Oct**

PAIR CHECK

- Could you guess which **figures** the report would need?
- Is the scope small enough for **3–5 figures** and 2,000 words?
- Is there a real ‘**however**’, or just a topic?

Using the literature: synthesis, not a grocery list

Schultz: describe the topic, not the papers. Organise sources around ideas, and show what is known versus what this report adds.

GROCERY LIST

“Smith (2001) studied low clouds. Jones (2005) looked at ENSO. Lee (2012) found that SST affects clouds. Brown (2018) also studied this.”

SYNTHESIS

“Subtropical low cloud is closely tied to the stability of the lower troposphere (Smith 2001; Lee 2012), so ENSO-driven changes in sea-surface temperature could alter it. Whether this happens in the same sense across the eastern Pacific is less clear (Jones 2005; Brown 2018), which motivates comparing two regions here.”

WHAT CHANGED

- The **idea** is the subject of each sentence; the citations sit in brackets as evidence.
- Sources are **grouped** by what they agree on, and the gap leads straight into **your** purpose.
- (The names here are placeholders. Cite real sources, and always cite **ERA5** and the **ONI**.)

Before Tutorial 4 · Thu 29 Oct

Two weeks, including reading week (no tutorial on 22 Oct). About 3½ hours of writing, plus notebook time.

3A · 45 min

Opening paragraph(s), ~250 words

- For your **leading** candidate question
- Context → 'however' → purpose statement
- No banalities in sentence 1

3B · 75 min

Literature synthesis paragraph

- Find **3 relevant sources** (at least 2 peer-reviewed)
- Write **one** ~150-word paragraph that synthesises them around one idea
- Full references in one consistent style

3C · 30 min

Three candidate questions

- Each: the question + "I find this interesting because..."
- + the **figure** that would answer it
- Bring all three on 29 Oct

NOTEBOOK STRAND

- **Sections 4–5** (+ 6 if timing interests you); try other cloud levels (VARIABLE).
- Optional: **methods_demos.ipynb §1–2**. Analysis slides: **04**.

READ · SCHULTZ (2009) ELOQUENT SCIENCE

- **§3.1–3.2** Titles
- **§4.6–4.7** Introduction; literature synthesis
- **§12.1–12.3** Why, how and what to cite